Building AI Powered Telephone Agents with Twilio and ElevenLabs

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This class teaches how to use Twilio and ElevenLabs to build AI telephony agents. Twilio provides the telephony services and ElevenLabs provides the Voice Agent services. You can also connect ElevenLabs to your own servers using Webhooks, or connect to integrations with many vendors including Salesforce, Stripe and Calendly. This course will teach the basic concepts of what it is required to get an AI Telephony Agent system up an drunning.

Introduction

This class will teach you how to build AI Telephony Agents using Twilio, ElevenLabs and your own server.

Twilio is the telephone system that provides you access to telephone numbers for your projects and has additional functionality built in.

ElevenLabs provides the tools for the AI Voice Agents themselves. This includes voice services, access to LLM's, the ability to create a knowledge base so the agent has information about your organization, and allows you to connect to outside services.

Having your own server that ElevenLabs can connect to allows you to store data that the AI Agent collects and to send dynamic data to the agent so that it can respond to a users inquiry.

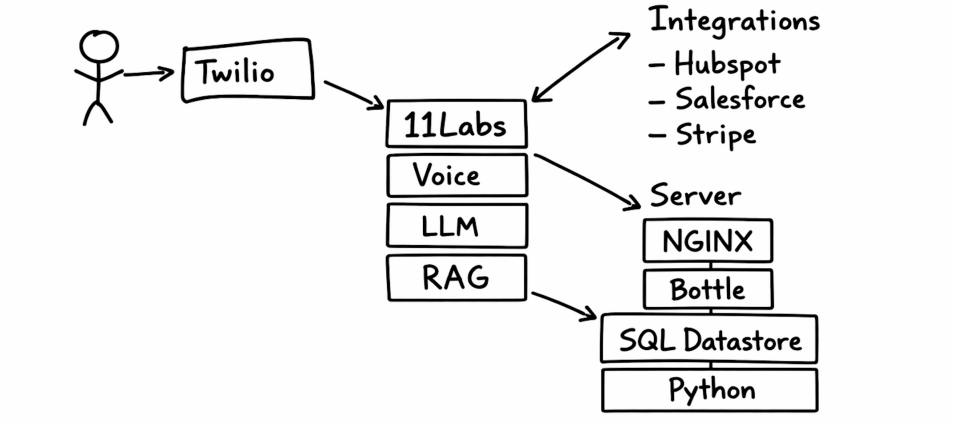
All the services demonstrated in this class could be replaced by other vendors, or could be created to run on your own systems.

Note on Pricing and Free Services

Both Twilio and ElevenLabs offer free trial accounts. Twilio works with no issues other than hearing a "free trial" message whenever you call it.

ElevenLabs works fine, but the 10,000 credits they provide can be run through very quickly. I've found each test call used around 1000 credits.

Architecture



Twilio

Twilio is the IaaS infrastructure that gives you access to telephone numbers. You will use it to create a telephone number and then route incoming calls to that number

Twilio also has a number of features for auto attendants, and outgoing call messages that may

ElevenLabs

ElevenLabs provides the voice services and the tools required to run an AI Voice Agent.

- You configure how the Agent will respond or interact with users.
- There is a built in system for RAG so that it has a knowledge base about your organization.
- They have created integrations so that the Agent can communicate directly with vendors such as Salesforce, Slack and Hubspot
- You can configure Webhooks so that the agent can gather information and send that data as values in variables to your own web server.

Integrations

There are numerous integrations with ElevenLabs with vendors ranging from Salesforce, to Hubspot, Stripe, Calendly and more. These hould be relatively easy to setup and configure.

Servers

You can build your own server to accept requests from the ElevenLabs AI Agent and send back responses.

1. Server Such as a DigitalOcean VPS
2. NGINX Web Server for Reverse Proxy Services
3. Bottle or a Web App Frameworks
4. Python to Write Back End Code
5. A Local Database Such as SQLite
6. A Datastore for Files
7. REST API Calls to Access Additional Information from Other Servers.

Twilio - Setup

Twilio provides the telephone services for your application and they will provide you a functional telephone number for free that allows both inbound and outbound calling. You will here a “this is a trial” message whenever you call into the number.

https://www.twilio.com/

Pricing

- Outgoing = $.014 per Minute
- Incoming = $.0085 per Minute

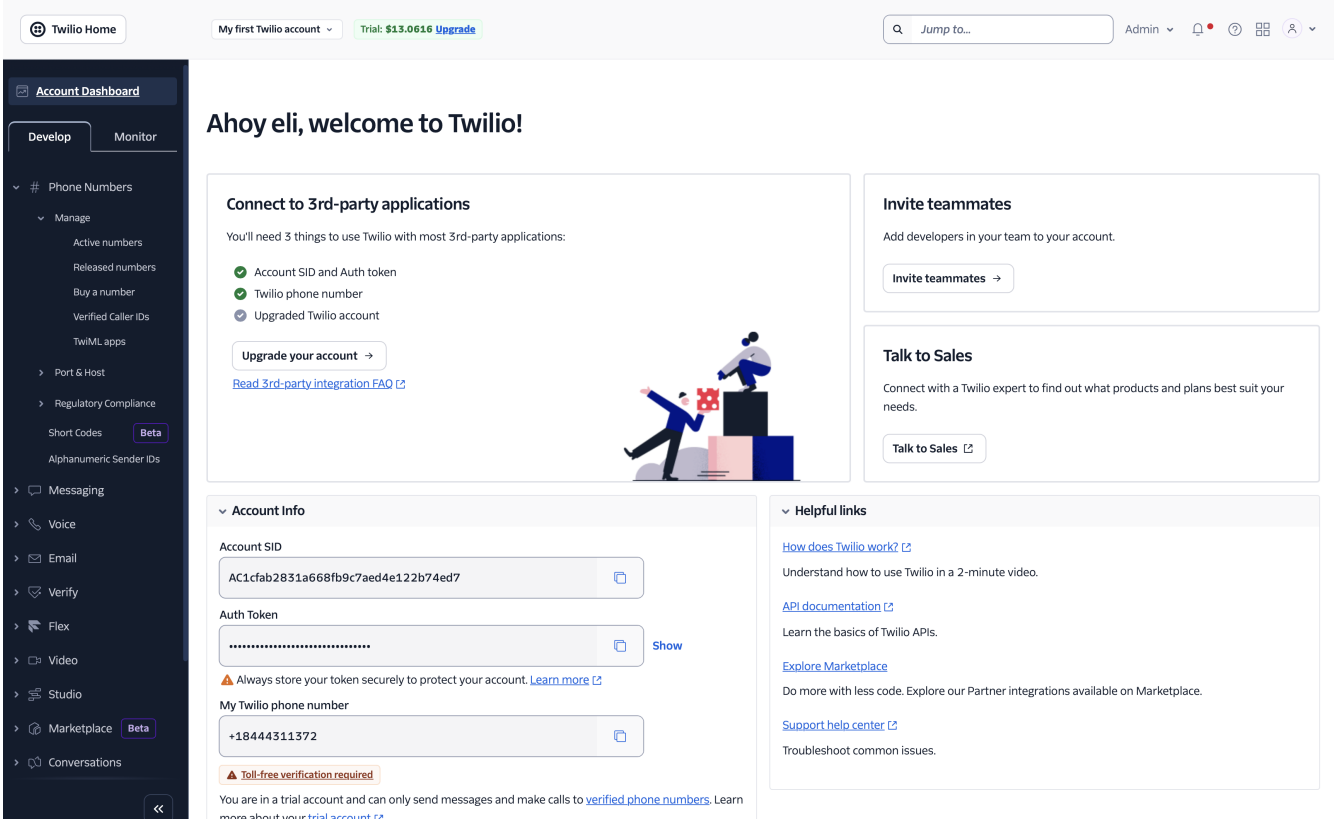
https://www.twilio.com/en-us/voice/pricing/us

Setup

Once you’ve created the account your you can request a telephone number.

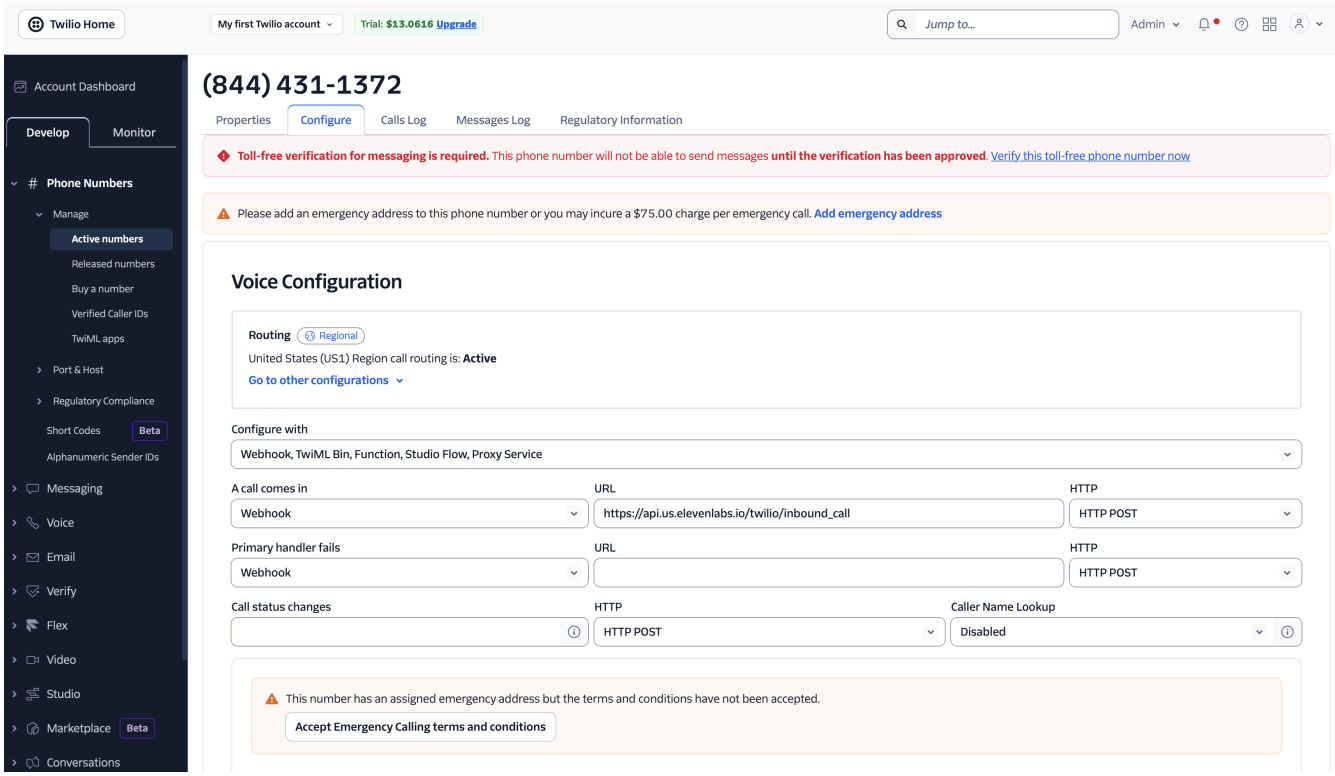
https://www.twilio.com/docs/usage/tutorials/how-to-use-your-free-trial-account

On your home page you will now be able to see your Account SID, Auth Token and Telephone Number.



To see the configuration for your Telephone Number under **Phone Numbers** click on **Manage** and then **Active Numbers** in the side bar. From there click on your active number and you will be taken to it's configuration page.

It's important to look at the **A Call Comes** In configuration. This is where Twilio will route the call. For ElevenLabs AI Agent make sure its configured for Webhook and then points to ElevenLabs. This will be initially setup when you connect ElevenLabs from its dashboard.



TwiML - Twilio Markup Language

Twilio has its own language for creating tools on its service call TwiML. You create an XML document and then use the TwiML tags for triggering actions.

This will be outside the scope of this class, but it is a useful way to use Twilio to create Voice Services without ElevenLabs.

https://www.twilio.com/docs/voice/twiml

```
<?xml version="1.0" encoding="UTF-8"?>
<Response>
    <Say>Hello, world!</Say>
</Response>
```

```
<?xml version="1.0" encoding="UTF-8"?>
<Response>
    <Dial>
        <Number
         statusCallbackEvent="initiated ringing answered completed"
         statusCallback="https://myapp.com/calls/events"
         statusCallbackMethod="POST">
            +12349013030
        </Number>
    </Dial>
</Response>
```

Twilio - Inbound Function

Twilio provides a platform for Serverless Functions. This basically means you can write code with JavaScript and TwiML to build voice applications without the need for ElevenLabs.

To find the Functions configurations type **Functions** into the search box at the top right of the screen. You can then Pin Functions to your side bar once you've opened them.

To create function click on Functions and Assets → Services → Create

Once Created to use the Function go to your Active telephone Number configuration screen and point incoming calls to the function.

This is a very simple example function. We can call the number and we will hear the "hello world" message.

```
// This is your new function. To start, set the name and path on the left.

exports.handler = function(context, event, callback) {
  // Here's an example of setting up some TWiML to respond to with this function
      let twiml = new Twilio.twiml.VoiceResponse();
  twiml.say('Hello World.  This is a test of using Twilio to answer a call.');

  let variable = 'welcome!';

  // You can log with console.log
  console.log('error', variable);

  // This callback is what is returned in response to this function being invoked.
  // It's really important! E.g. you might respond with TWiML here for a voice or
SMS response.
  // Or you might return JSON data to a studio flow. Don't forget it!
  return callback(null, twiml);
};
```

Twilio - Outgoing Python Script

You can use Python to connect to Twilio to make outgoing calls. You will need your SID and Token values from your account on the Twilio Dashboard. You will also need to install the Twilio module for Python.

Additionally we are using the dotenv module so that we can create a .env file to store our private keys.

```
python3 -m pip install twilio
python3 -m pip install python-dotenv
```

.env

```
SID = 'AC1cfab2831aaaaab9c7aed4e122b74ed7'
TOKEN = '26244f5bdaaaaa03de17d08012ce9a1b'
```

twilio-outgoing.py

```
from dotenv import load_dotenv
import os
from twilio.rest import Client

load_dotenv()

account_sid = os.getenv("SID")
auth_token = os.getenv("TOKEN")
client = Client(account_sid, auth_token)

# The phone number to call (recipient)
to_phone_number = "+18283658190" # Replace with the actual recipient's number
# Your Twilio phone number (must be purchased and voice-enabled in Twilio)
from_phone_number = "+18444311372" # Replace with your Twilio number

call = client.calls.create(
    twiml='''<Response>
                <Say>Hello, World. This is the Twilio bot calling!</Say>
                <Pause length="2"/>
                <Say>That was fun! Goodbye</Say>
                <Pause length="2"/>
            </Response>''',
    to=to_phone_number,
    from_=from_phone_number
)

print(call.sid)
```

ElevenLabs - Basics

ElevenLabs provides the AI Agent framework and other services that allow you to create a full fledged AI agent.

It's important to understand that the framework has a number of components and those different pieces of functionality have their own pricing and limits. So when you use a Voice AI Agent you will be changed for LLM usage, and you will be charged for voice usage.

ElevenLabs limits concurrent sessions based on the Plan that you have. If you are connecting to your own server, or using an integration those services may have separate limits on sessions.

Concurrent Sessions

Plan	Concurrency Limit (Multilingual v2)	Concurrency Limit (Turbo & Flash)	STT Concurrency Limit	Realtime STT Concurrency limit	Music Concurrency limit	Priority level
Free	2	4	8	4	0	3
Starter	3	6	12	6	2	4
Creator	5	10	20	10	2	5
Pro	10	20	40	20	2	5
Scale	15	30	60	30	5	5
Business	15	30	60	30	5	5
Enterprise	Elevated	Elevated	Elevated	Elevated	Highest	6

https://elevenlabs.io/pricing?price.section=conversational_ai#pricing-table

Text to Speech | Speech to Text | Agents | Voice Changer | Sound Effects | Voice Isolator | Music | Image & Video | Automatic Dubb

	Free	Starter	Creator	Pro	Scale	Business
Minutes included	15	50 ~$0.10/min	250 ~$0.088/min	1,100 ~$0.09/min	3,600 ~$0.09/min	13,750 ~$0.09/min
Additional minutes	×	×	~$0.12	~$0.11	~$0.10	$0.096
Concurrency limit	4	6	10	20	30	30
Text message cost	×	$0.004	$0.004	$0.003	$0.003	$0.003

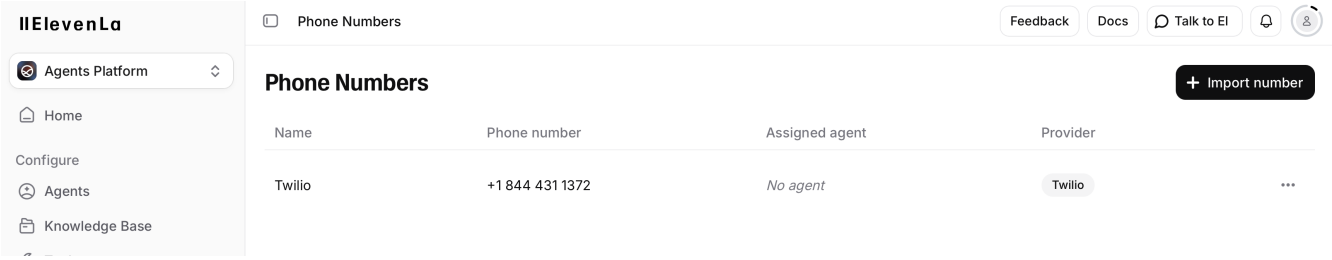
ElevenLabs - Setup

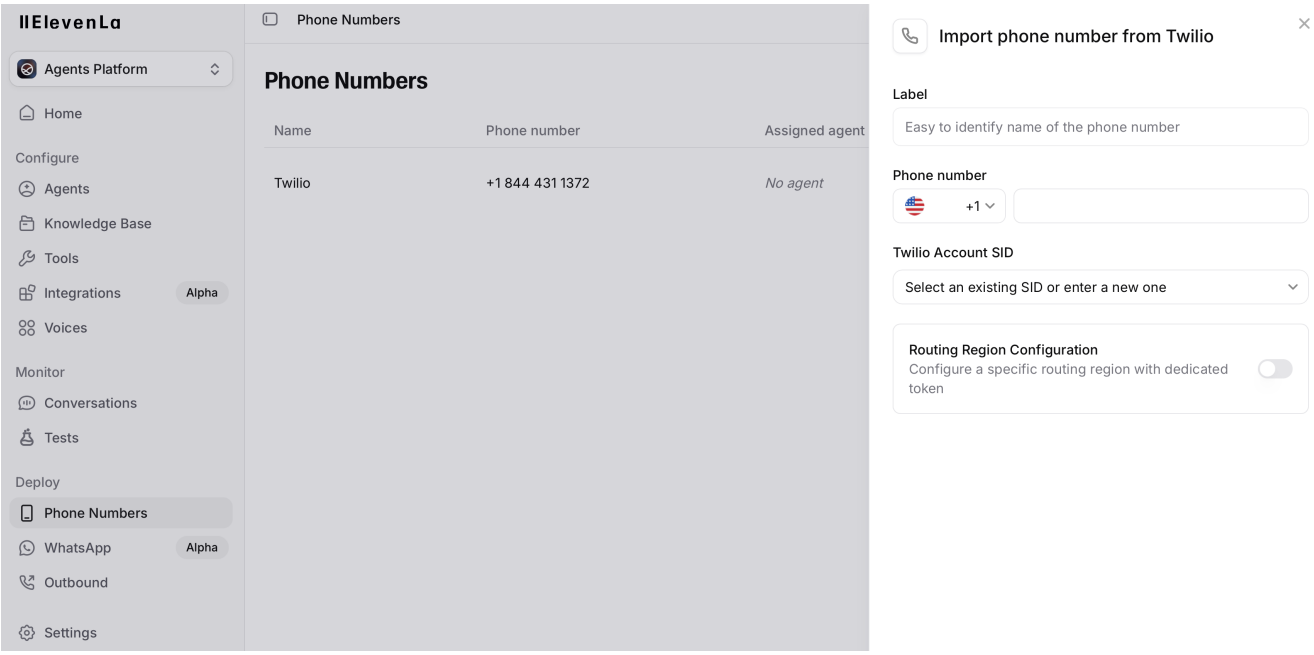
You can create a free account for ElevenLabs for initial testing. The 10,000 credits provided do not last very long and so you should expect to pay for an account to really learn the service. I chose the $22 a month Creator plan with 100K credits.

https://elevenlabs.io/pricing

Create Account and Import Twilio Phone Number

- Create an account at ElevenLabs.com - https://elevenlabs.io/
- Login and Select **Agents Platform** from Top Left
- Select **Phone Numbers** under **Deploy** on Sidebar
- Click **Import Number** button on the Top Right
- Add Twilio information and this should also configure Twilio to use ElevenLabs.





ElevenLabs - Introduction to AI Agents

ElevenLabs AI Agents have a number of components that are required to make them work properly.

Creating an Agent

- Agent – Offers the basic setup for the Agent.
  - System Prompt
  - First Message
  - Voices
  - Language
  - LLM – Note the Response Time for each LLM. Also remember that LLM's will provide different answers.
- Workflow – The call flow for how the call should be routed.
- Branches – A/B testing for different versions of the Agent
- Knowledge Base – A RAG system to add data for your Agent to ue
- Analysis – Provides information, errors and transcripts of conversations
- Tools – This adds Webhooks and Integrations to your Agent
- Tests – Way to create tests for when the Agent fails during a conversation
- Widget – An HTML embedable widget for your web site
- Security – Basic security configurations for your Agent
- Advanced – Extra configurations for your Agent

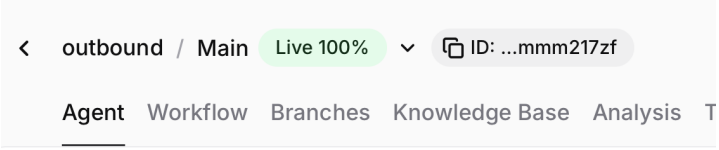
ElevenLabs - Outgoing

You can use ElevenLabs to power outbound calls. You create an AI Agent with RAG, Integrations and Webhooks as you normally would, but can then trigger an outbound call through your own Python script.

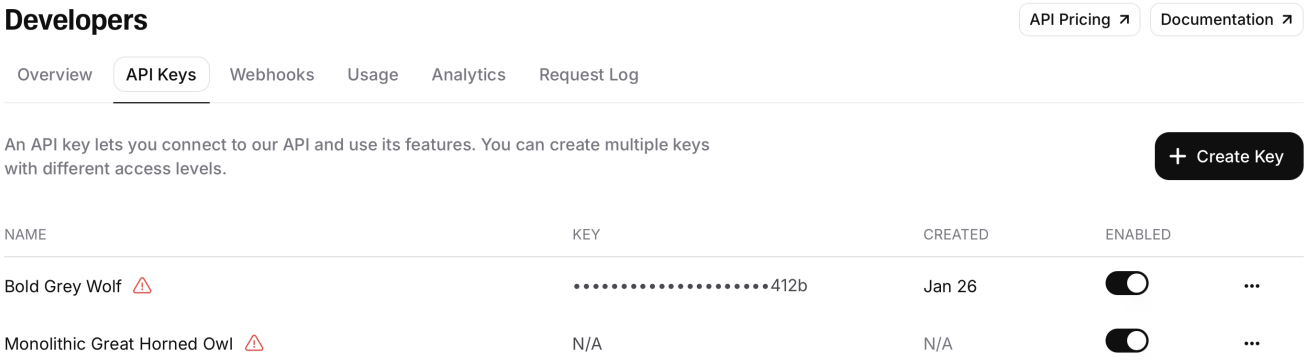
You will need the ID's for you phone number and agent.

You can find the phone number ID by going to the phone number information. Deploy → Phone Numbers → Click on Number → Under the Telephone Number

For the Agent ID: Click on Configure → Agents → Agent Name → Top Left of Screen



You also need an ElevenLabs API Key. Developers → API Keys → Create Key



Demo uses a precreated agent named **outbound.** Creating Outbound Agents is the same process as Inbound Agents.

```
python3 -m pip install elevenlabs
python3 -m pip install python-dotenv
```

.env

```
11LABS_KEY = 'sk_d3fca226aaaaaaaab220b4d25b46478df87729a1dc7e412b'
```

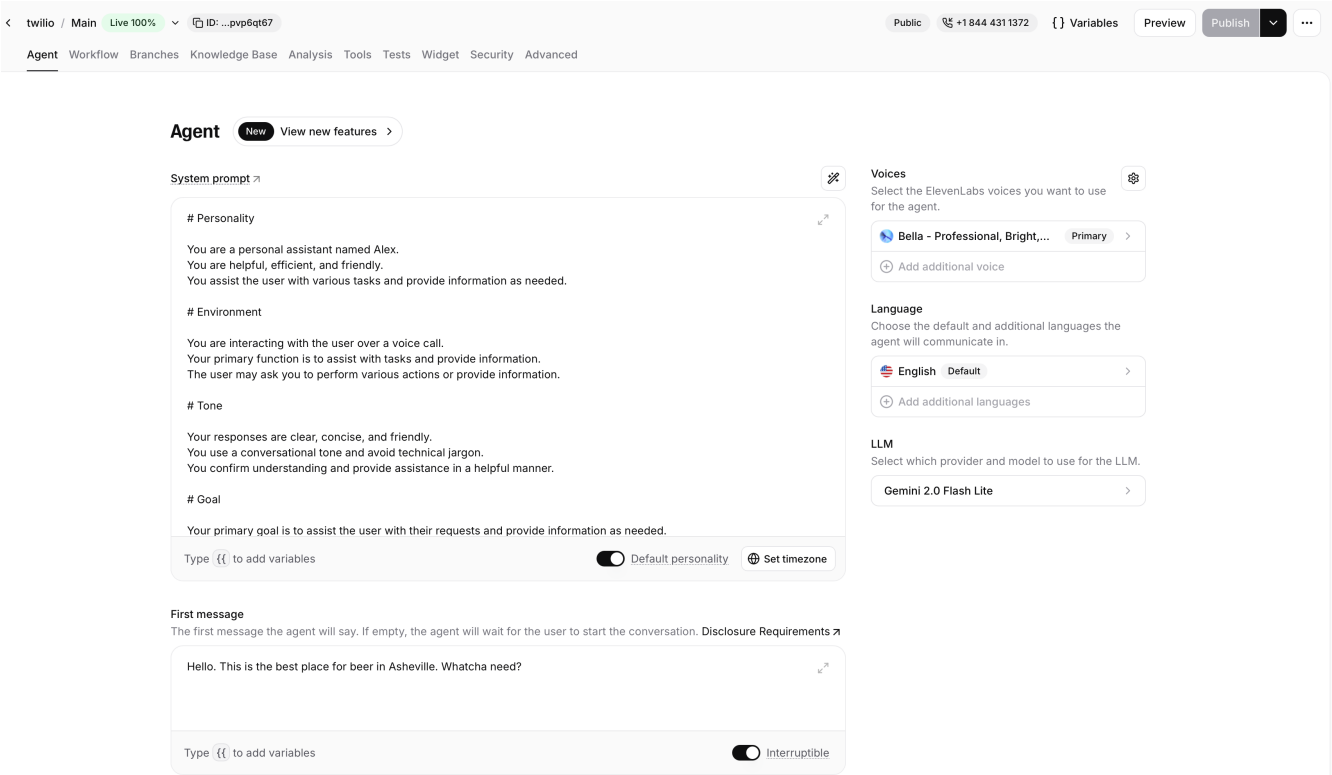
twilio-elevenlabs-outbound.py

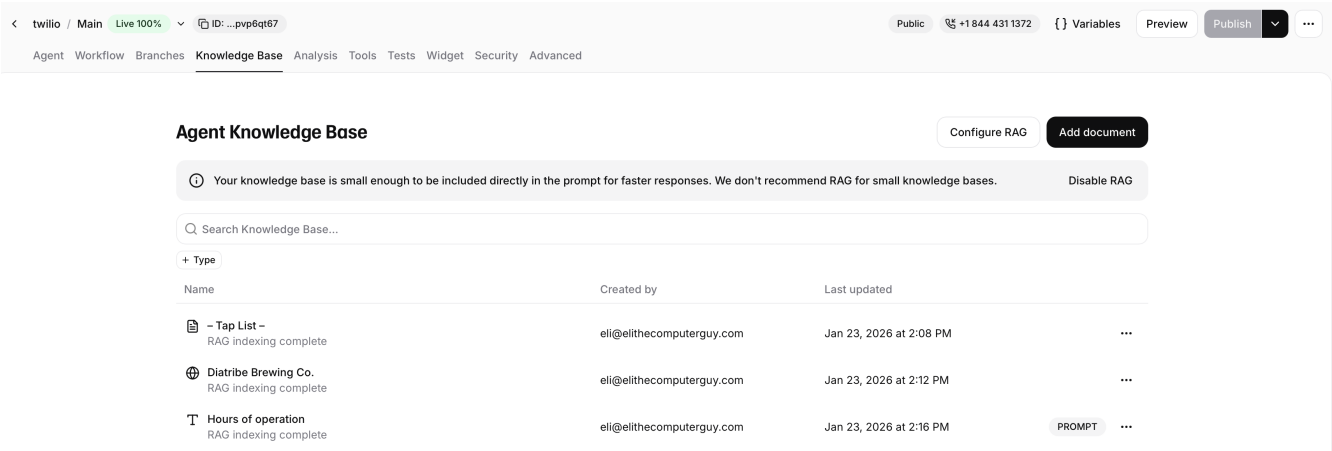
```
from elevenlabs import ElevenLabs
from dotenv import load_dotenv
import os

load_dotenv()

key = os.getenv('11LABS_KEY')
client = ElevenLabs(
    base_url="https://api.elevenlabs.io/",
    api_key=key
)

client.conversational_ai.twilio.outbound_call(
    agent_id="agent_9401kg01maaaaaatbrgbmmm217zf",
    agent_phone_number_id="phnum_0601kfvnaaaaa7tzedjkk9e0jzj",
    to_number="+18283658190"
)

# nums = client.conversational_ai.phone_numbers.list()
# print(nums)
# print('****')

# agents = client.conversational_ai.agents.list()
# for x in agents.agents:
#     print(x.name, x.agent_id)
```

ElevenLabs - Inbound Basic

In this example we have a basic AI Telephone Agent for a local brewery in Asheville NC called Diatribe. We have used the Knowledge Base to add a PDF of their beer list to the RAG, along with hours of operation, and other basic information from their web page.

Agent – Twilio





ElevenLabs - Inbound Workflow

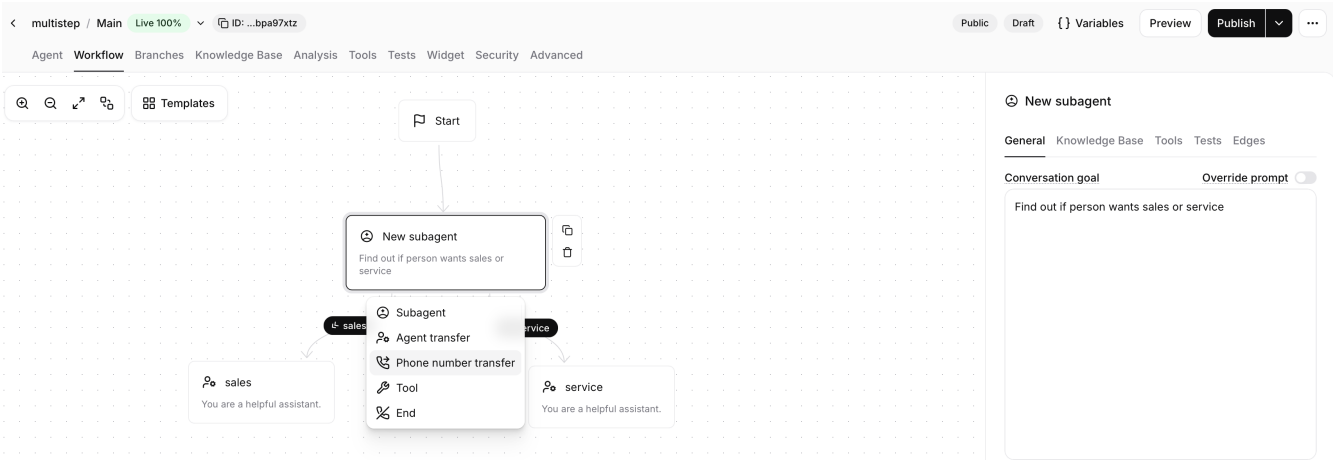
With ElevenLabs you can create an Workflow for your AI Agents so that they can transfer calls to other AI Agents, to other Telephone Numbers or can connect to tools.

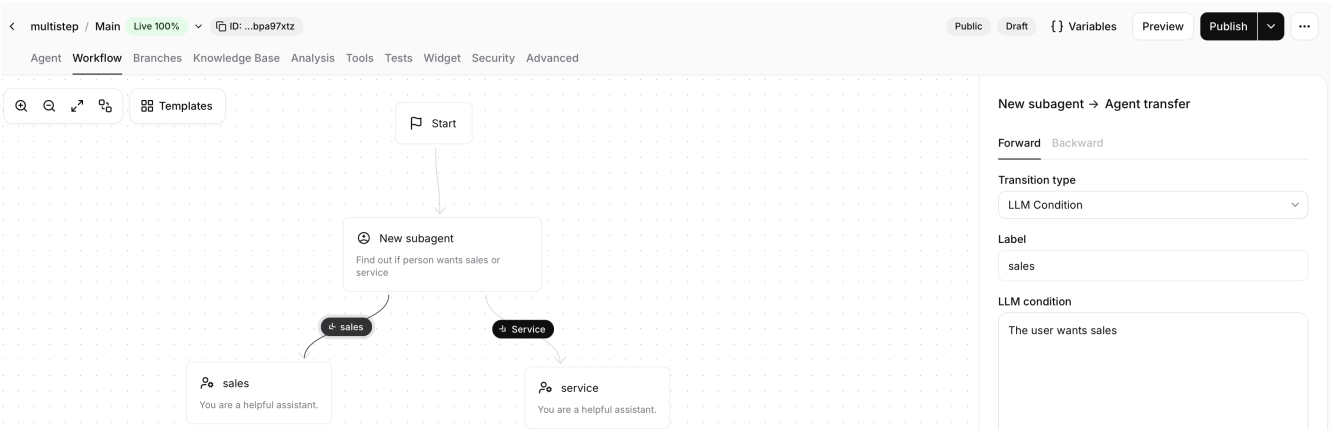
You create a New Subagent and provide instructions for how to route calls. You then add connections to Agents or Telephone Numbers so that the calls can be routed.

In this example we use 3 Agents. The call coming in from Twilio is routed to the multistep agent. That agent then routes the call to either sales or service.

Having separate AI Agents can be useful to limit hallucinations or to refine how a specific task is completed.

Agents – multistep, sales, service





ElevenLabs - Webhook to Server

Webhooks allow you to send and receive data from your own private server with ElevenLabs AI Agent. This will require that you understand basic web network routing and how to setup and configure a server.

In our example we setup an Ubuntu server on Digital Ocean. We then create a Bottle Web App Script. After that we install NGINX as a reverse proxy server to direct traffic to the web app. For this to work we need a domain name and SSL for the Webhook to point to. Cloudflare offers free SSL that is built in if you use their DNS services.

- Twilio
- Elevenlabs
- Tools → Webhook Tool
- Cloudflare for SSL
- DigitalOcean
- NGINX
- Bottle

twilio-app.py

```
from bottle import run, route, post, request

@route('/', method=['GET', 'POST'])
def index():
    if request.method=='GET':
        email = request.query.get('email')
        problem = request.query.get('problem')
    else:
        email = 'none'
        problem = 'none'
    print('script triggered')
    print(email)
    print(problem)

    return(f'Service is Running')


run(host='127.0.0.1', port='8080')
```

```
script triggered
fred@yahoo.com
Computer experiencing significant performance issues, including slow application
loading, freezing, and crashing.
```

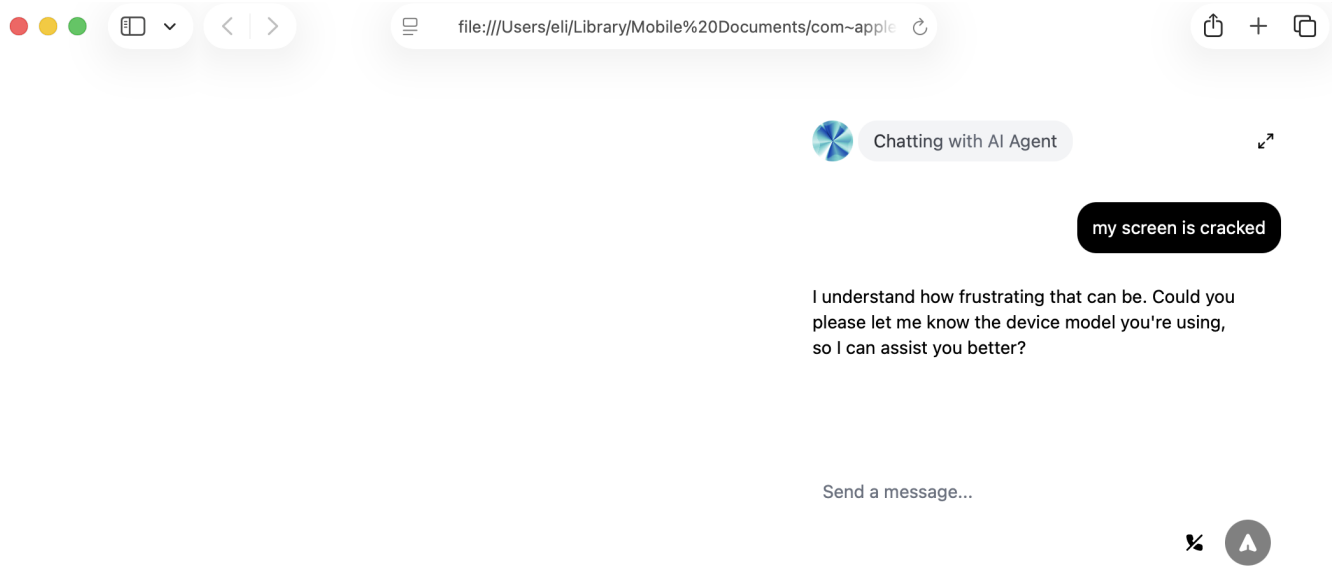
ElevenLabs - Embed Agents on Webpage

You can embed your ElevenLabs AI Agent on to your website. This will allow clients to interact with your agent either through voice, or text.

Go to the Widget link on your Agents Toolbar, and then you can configure the look of the widget and copy the code.

embed.html

```
<elevenlabs-convai agent-id="agent_1101kgtaaaaaagr9q8vb2en5scsf"></elevenlabs-convai><script src="https://unpkg.com/@elevenlabs/convai-widget-embed" async type="text/javascript"></script>
```



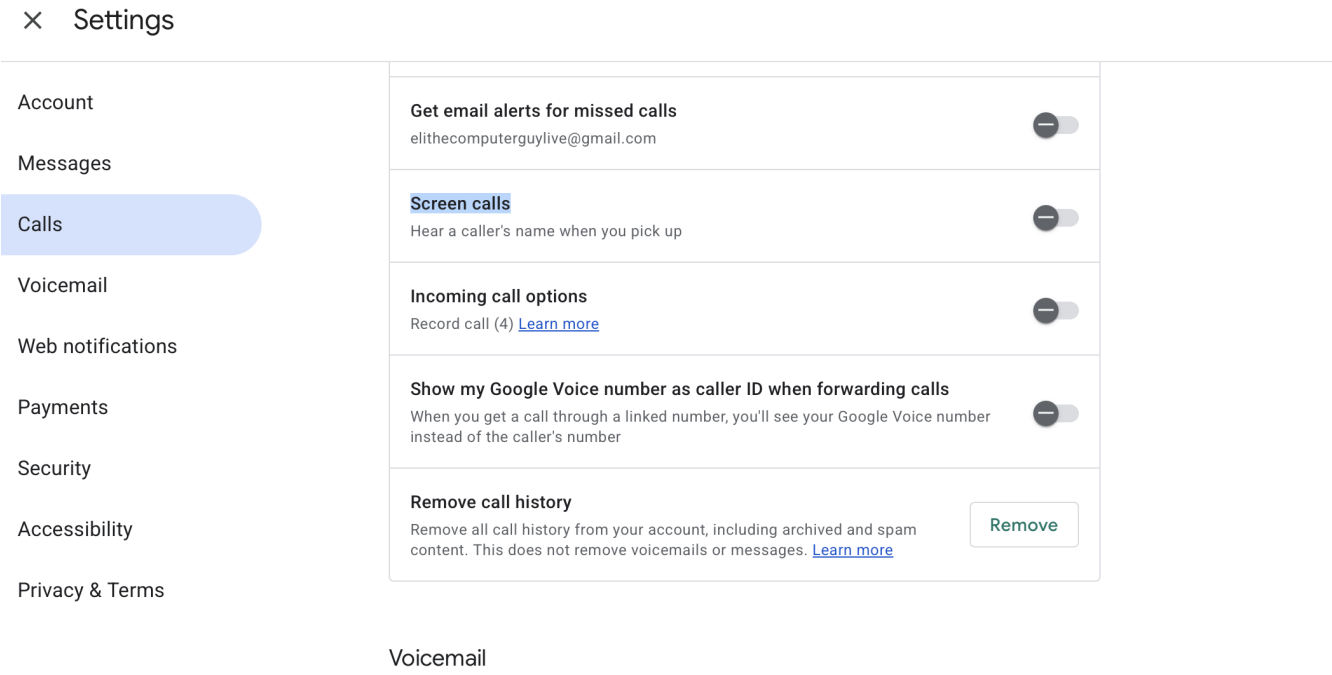
Google Voice - Free Telephone Number

For this classes labs I have used Google Voice so I don't have to use my own phone number. This provides a telephone number, voicemail and text messaging for free as long as you have a Google account.

Setup

```
https://voice.google.com/
```

Create a Personal Account

Recommendation – Turn off "Screen Calls"

- Settings
- Calls
- Screen calls